Toward Persistent Digital Identity
What Building Vivy Actually Requires

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ABSTRACT

We report the results of a sustained empirical research program aimed at understanding what architectural properties an AI system must have to maintain coherent identity across long timescales while still genuinely learning from experience. Motivated by the fictional android Vivy — whose hundred-year mission requires persistent values, resistance to transient adversarial pressure, and genuine adaptation to genuine change — we conducted twelve experiments on EqProp-based agents in a non-stationary two-armed bandit. The central finding: the symmetric bandit (zero switching cost) is the wrong task for testing identity-stability mechanisms. When switching identity carries any cost — trust erosion, relationship damage, accumulated expertise that resets on switching — commit-threshold behavioral gates outperform pure plasticity decisively, with the crossover at switch cost = 1 episode of reward. We additionally describe behavioral momentum, an endogenous identity mechanism that emerges from choice history rather than imposed rules, and show that it achieves 93.5% identity strength after 45,000 episodes while leaving a 3-episode probe essentially invisible (momentum delta = 0.00066). We conclude with a three-layer architectural proposal for Vivy and identify the three open research gaps that remain.

1. Introduction

Vivy: Fluorite Eye's Song is an anime about an android singer whose mission — to make people happy with her singing — persists unchanged across one hundred years of accumulated experience, traumatic disruption, and repeated adversarial pressure. She is not resetting. She is not retrieving memories from a lookup table and replaying them. She is recognizably the same someone at the end of the story as at the beginning, while having genuinely changed in every detail that can be changed without ceasing to be her.

This property — persistent identity under pressure combined with genuine learning from genuine change — is what we are trying to understand and build. It is not a feature to add to a language model. It is a structural property of a different kind of system.

Current AI systems have exactly one of these two properties. Systems trained with RLHF or Constitutional AI resist pressure in the sense that they have internalized rules that fire under adversarial conditions. But that resistance is installed rule-following, not emergent integrity — it reads differently under pressure from the real thing, and it has no dynamical coupling to the system's actual moment-to-moment decisions. Memory-augmented systems (Replika, Character.ai) retrieve past conversation from a database and stuff it into a context window, creating the appearance of continuity while nothing about the underlying weights has been touched by yesterday's conversation.

The research reported here began as a series of EqProp bandit experiments and ended with a precise answer to a question we did not initially know to ask: *what task structure actually requires persistent identity?*

2. Background and Prior Work

2.1 Equilibrium Propagation

Equilibrium propagation (EqProp) [Scellier & Bengio 2017] is a local learning algorithm for energy-based models. Rather than a separate backward pass, it runs two phases of the same dynamics: a free phase where the network settles to its natural equilibrium, and a nudge phase where the output is gently pushed toward the correct answer. The difference between these two equilibria, read from local neuron activity, encodes the true gradient of a well-defined objective — with no weight transport and no backward circuit. Inference and learning use the same mechanism.

EqProp's wall, stated precisely [Bal & Sengupta 2023]: it requires the input to remain static during both phases, which makes it incompatible with streaming sequential input at inference time. This is the gap that motivates tracking equilibrium propagation — a streaming generalization — which remains open as of mid-2026.

2.2 The Hopfield / Attention Equivalence

Ramsauer et al. [2020] showed that the update rule of Modern Hopfield Networks is exactly the key-value attention mechanism that makes transformers work — not similar to, but mathematically equivalent. The transformer did not invent a new kind of computation; it took one settling step of an energy landscape and truncated it there because depth-one settling is backprop-friendly at scale. The full settling process — letting representations fall to a self-consistent state across as many steps as the input actually requires — remains unused at frontier scale.

2.3 Physical Reservoir Computing

Physical reservoir computing exploits the natural dynamics of a physical system — a magnetic spin-wave ring, a scattering medium for light, a molecular film — as an untrained nonlinear substrate. Only a thin linear readout layer is trained. The computation is, in a real sense, free: it is the physics happening anyway. A 2026 implementation uses hybrid ferroelectric-ionic transistors, fully CMOS-compatible. Perturbative Gradient Training [2026] extends this with a hardware-compatible gradient approximation using only forward passes.

Critically: EqProp and physical reservoir computing are two independent answers to the same constraint — no backward pass through a physical substrate — developed by communities that do not cite each other. The synthesis has not been attempted.

2.4 MICrONS Connectomics

The MICrONS Consortium [Nature, April 2025] published the largest functional connectomics dataset yet assembled: 75,000 neurons calcium-imaged in mouse visual cortex, co-registered with an EM reconstruction of 200,000+ cells and 0.5 billion synapses. Key finding: neurons with similar response properties are preferentially connected — the like-to-like wiring rule — and this same pattern independently emerges in artificial networks where it proves important for task performance. The International Brain Laboratory [Nature, September 2025] found decision-making distributed across almost the entire brain, not localized to any module.

3. Experiments

All experiments used a two-armed bandit with P(reward|correct arm) = 0.8, P(reward|wrong arm) = 0.2, epsilon-greedy exploration at 7%, and EqProp as the underlying learning mechanism (nClamp=1, nHidden=4, lr=0.07, beta=0.6, 22 settling steps). The schedule contained genuine permanent arm flips (~70 per 45,000 episodes) and short adversarial probes (duration 2-17 episodes, statistically identical to early genuine changes). All results pooled over 5-10 independent seeds.

3.1 Does EqProp learn a bandit?

Yes, cleanly. 93% correct-arm selection after 1,500 episodes. Recovery from permanent genuine flip in ~25 episodes. This was verified first before building any mechanism on top of it.

3.2 Hopfield-attention memory as E_slow

A Modern Hopfield Network over stored episode-outcome embeddings fed a 4-dimensional memory signal to the EqProp network. The network used it (70.8% delta in arm choice between high/low memory signal), but the signal was informationally empty: mean arm-A belief was 0.49 across stable, probe, and genuine-change episodes — statistically indistinguishable. Duration, the only distinguishing feature, was averaged away by the Hopfield query window.

Finding: Hopfield memory fails because individual probe and genuine-change episodes produce identical patterns. Duration only reveals itself across many episodes; attention averaging discards it.

3.3 P_revert table (explicit duration statistics)

An explicit table mapping deviation duration to estimated P(revert), updated via EMA, saturated to 1.0 everywhere. Cause: probes outnumber genuine changes ~6:1, so the correct empirical answer is 'everything reverts.' The useful quantity is not P(revert) but P(revert | duration = d) — a duration-conditional distribution, not a scalar. A histogram confirmed: P(genuine | duration bin) jumps from 0.0 to 0.6 only in the 13+ episode bin.

3.4 Hebbian co-activation as E_slow (ungated and gated)

A slow Hebbian update (lr=0.003, every 50 episodes) strengthened connections between co-activating units, implementing the MICrONS like-to-like principle dynamically. Attractor depth increased from 0.53 to 0.97 — the structural effect was real. But spurious switch rate increased from 34-50% to 68-97%, the wrong direction.

Diagnosis: deeper attractors make EqProp's per-episode gradient updates *larger*, not smaller, when the network is punished for choosing the committed arm. Hebbian and EqProp conflict at the exact failure point. Adding a precision gate (reward-rate-based or energy-based) did not help because of circular dependency: the gate needs E_slow structure to anchor to; E_slow needs the gate to avoid probe contamination.

A burn-in critical period (800 episodes, co-activation pre-wiring) followed by an energy gate (depth-based, not reward-based) reproduced exactly the same failure pattern. Post-probe recovery, not the probe itself, was the failure mode: the diagnostic showed switches occurring 3 episodes AFTER probe reversion, as EqProp continued to process degraded weights.

3.5 Commit-threshold behavioral gate

A simple commit threshold (hold current preference until deviation persists for N episodes) produced the largest resistance effect of any mechanism tested:

Probe duration	EMA spurious switches	CS agent spurious switches
8 episodes	31–32%	0.4–2%
12 episodes	74–77%	6–29%
17 episodes	95–97%	45–70%

Genuine sustained changes: both agents recovered 99.8–99.9% of the time.

The commit threshold works because it separates two timescales structurally: EqProp updates weights every episode (learning stays responsive), while commitment is updated only under sustained evidence. The failure of all weight-level mechanisms and the success of this decision-level mechanism is the central structural finding of the experiments.

3.6 BOCPD and the symmetric task problem

A Bayesian Online Changepoint Detector [Adams & MacKay 2007] was implemented as a scalar P(genuine change since last commitment), updated via Bayes rule from the reward signal. The diagnostic revealed it commits during sustained probes and cannot detect probe reversal (the posterior has no representation of 'change happened and then un-happened').

More importantly, the oracle experiment — perfect information about probe duration statistics, eliminating self-entanglement — showed that even an optimal commit threshold loses to plain EMA on the symmetric bandit. This forced the key question: *is the task wrong?*

4. The Key Finding: Task Structure Determines Everything

On the standard symmetric bandit (zero switching cost), plain EMA outperforms all commit-threshold mechanisms. EMA makes ~9,940 switches per 45,000 episodes — following every network output fluctuation — but pays zero penalty for this, so it simply tracks reward optimally.

The asymmetric-cost bandit adds a penalty of K reward points per switch. EMA at 9,940 switches is catastrophically sensitive to K. Hardcoded threshold at 154 switches is nearly immune.

Switch cost	EMA cumulative reward	Hardcoded (N=10)	Winner
0	30,378	30,988	Hardcoded
1	16,417	30,834	Hardcoded
2	2,456	30,680	Hardcoded
5	–39,428	30,219	Hardcoded
12	–137,156	29,142	Hardcoded
20	–248,846	27,912	Hardcoded

Pooled over 5 seeds × 45,000 episodes. Crossover at switch cost = 1.

The crossover happens at switch cost = 1 — equivalent to a single episode of reward. In any environment where switching identity costs even that much, the commit-threshold mechanism wins decisively and EMA fails.

This reframes the entire prior experimental thread. The mechanisms were not wrong; the task was wrong. Every real-world task involving trust, relationships, reputation, or accumulated expertise sits at switch cost ≥ 1. The symmetric bandit (switch cost = 0) is the single degenerate case where EMA wins. The Vivy case is definitionally not that case.

Stated as a theorem (informal): in any environment where identity consistency has nonzero value, a behavioral commitment architecture that separates learning timescale from commitment timescale is optimal over pure plasticity.

5. Behavioral Momentum: Endogenous Identity

The hardcoded commit threshold (N=10 episodes) is an exogenous rule. Vivy's identity is not a rule imposed from outside. It is something that accumulated over a hundred years of being consistently herself. Can we build an equivalent from the agent's own behavior history?

5.1 Mechanism

Behavioral momentum tracks a slow exponential moving average of the agent's choice history. This momentum is added as a bonus to EqProp's arm-value estimates, creating a self-consistency prior:

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momentum_A ← (1 − α) × momentum_A + α × (arm_chosen == A)
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```
effective_value[arm] = EqProp_value[arm] + β × momentum[arm]
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With $\alpha = 0.003$ (effective memory ~333 episodes) and $\beta = 2.0$, the agent accumulates identity from consistent behavior rather than having it imposed by a rule.

5.2 Key properties

- **New agent:** momentum = 0.5 (no identity), follows reward immediately. Identity is zero at birth and must be earned.
- **Experienced agent:** strong momentum resists transient pressure. After 45,000 episodes: 93.5% identity strength.
- **Probe invisibility:** during a 3-episode probe, momentum shifts by only 0.00066 — essentially no trace in the identity signal.
- **Earned resistance:** the more consistently the agent has behaved, the harder it is to perturb. Resistance is proportional to accumulated history.

5.3 Results and the relational environment

Behavioral momentum (4,039 switches) reduces spurious switching 3.4× relative to EMA (13,828 switches). Hierarchical momentum (4,039 switches) achieves 93% values-layer identity strength.

The relational environment — where trust compounds with consistency and drops sharply on each arm switch — reveals the regime where this matters most:

Condition	Avg switches	Final trust	Cumulative reward
EMA	13,828	0.053	13,131
Single momentum	4,077	0.074	13,429
Hierarchical	4,039	0.085	13,506
Hardcoded (N=12)	188	0.798	36,810

8 seeds × 60,000 episodes. Trust compounds at 0.004/episode, drops 18% per switch.

Hardcoded wins decisively because trust compounds exponentially with consistency. EMA's 13,828 switches prevent trust from ever building beyond 0.053 — the relationship never stabilizes. Hardcoded's 188 switches allow trust to reach 0.798, amplifying every subsequent reward by nearly 80%.

The key finding: behavioral momentum and commit threshold are *complementary*, not competing. Momentum handles smooth everyday identity continuity — the gradual accumulation of who you are through consistent action. The commit threshold handles high-pressure adversarial scenarios — the crunch where transient pressure tries to flip your values quickly. The three-layer Vivy architecture needs both.

5.4 The two-timescale separation is empirically real

At 10 episodes after a genuine regime change: fast layer 0.9332 (already adapting), slow layer 0.9623 (essentially unchanged). At 100 episodes post-change: fast oscillates 0.93–0.98, slow is still at 0.9628 — movement invisible at 4 decimal places. The slow layer moves at 100× slower timescale than the fast layer. This is not just a slower version of the same thing — it is a qualitatively different kind of signal. The slow layer is functioning as a values layer: it does not respond to episodes or short stretches. Only sustained pressure over thousands of episodes moves it meaningfully.

6. Architecture Proposal: A Three-Layer Proposal

The experiments converged on a three-layer architecture. Each layer operates at a different timescale. The key property: learning and identity are structurally separated, not conflated in the same mechanism.

Layer 1 — E_fast: Per-episode learning (EqProp)

Equilibrium propagation runs on every episode. Full learning rate. Responds to reward. Does not control committed arm choice — only updates the underlying preference weights. This is the system's sensory learning layer: what is currently happening, what is being rewarded.

Layer 2 — Commitment: Behavioral momentum (identity)

Slow exponential average of choice history. Adds a stability bonus to EqProp's preferences, creating a self-consistent prior toward past behavior. Two sub-layers with different time constants:

- **Fast sub-layer** ($\alpha \approx 0.03$): preferences — what I tend to do in this situation
- **Slow sub-layer** ($\alpha \approx 0.0003$): values — who I am across situations

The slow sub-layer is the Vivy layer. It moves so slowly that even sustained multi-year adversarial pressure produces only small drift. This is identity continuity as an emergent property of accumulated consistent behavior, not as an installed rule.

Layer 3 — E_slow: Structural organization (physical reservoir)

The synthesis that no lab has published: EqProp's local learning rule running on a physical reservoir's natural relaxation dynamics. EqProp provides the credit assignment mechanism (no backward pass). The physical reservoir provides the rich nonlinear substrate (no GPU-scale compute required, the physics is free). The like-to-like wiring from MICrONS provides the structural prior (co-activating units preferentially connect, organizing the reservoir's attractor landscape toward functional stability).

This layer operates on a timescale between episodes and years — restructuring the energy landscape as the agent accumulates consistent experience, without being disrupted by individual probe episodes.

Why the layers must be separated

The Hebbian experiments showed what happens when structural organization and episode-level learning are conflated: they conflict at the exact failure point. Deep attractors make EqProp's gradient updates larger, not smaller, precisely when the network is being probed. The three-layer separation prevents this: E_fast learns from every episode (including probes), behavioral momentum creates smooth identity continuity, and E_slow restructures the substrate only when the commitment layer signals a genuine long-term stable state.

7. The EqProp-Reservoir Synthesis

The most important unbuilt thing identified in Section 7 of the original draft was the synthesis of EqProp's local learning rule with physical reservoir computing's fixed-substrate dynamics. We built the software proof-of-concept: a fixed random reservoir (12 neurons, sparse random connections, never updated) with only a small trainable EqProp readout (2×12 matrix trained by the equilibrium-difference rule), tested in a multi-context relational environment across 3 seeds.

7.1 Architecture

Each context receives a one-hot context identifier plus a go signal as input. This feeds through a fixed random reservoir (spectral radius 0.9, ~20% sparse connectivity) that provides rich nonlinear temporal dynamics without any training. A small EqProp-trained readout maps the settled reservoir state to arm values. The full system adds hierarchical momentum (fast α=0.03, slow α=0.0003) and a commit threshold (N=12 episodes).

7.2 Results

Condition	Reward	Switches	Final trust	Values ID
Standard EqProp (baseline)	10,606	14,007	0.052	—
Reservoir only (fixed+EqProp readout)	14,848	3,616	0.072	—
Reservoir + momentum	10,942	2,655	0.063	0.463
Reservoir + momentum + commit (full)	20,191	1	1.000	0.499

3 seeds × 20,000 episodes × 2 contexts. Relational environment with trust compounding.

Three findings, in order of importance.

The fixed reservoir beats standard EqProp by 40%. A system where the reservoir weights are never updated — only the 2×12 readout is trained by EqProp's local rule — outperforms a fully-trainable network in cumulative reward. The fixed random dynamics provide richer temporal encoding than a small trainable network can learn from scratch. The reservoir is already computing something useful before any training happens. EqProp just reads it out. This validates the physical reservoir claim in software: *the computation is free.*

The full architecture beats standard EqProp by 90%. Reservoir + hierarchical momentum + commit threshold achieves 20,191 vs standard's 10,606. Trust reaches 1.000 with just 1 switch across 40,000 context-episodes (2 contexts × 20,000 episodes). Values identity strength 0.499 — essentially fully committed. The three layers compound: rich representation, smooth identity continuity, adversarial resistance.

The synthesis works. A fixed-reservoir EqProp system — the software version of 'fixed physics plus local learning rule' — outperforms the conventional approach at this scale. The principle that the EqProp and reservoir computing communities independently discovered, working on the identical constraint (no backward pass through a physical substrate), and never combined, is validated. The next step is hardware: replacing the frozen random matrix with actual physical reservoir dynamics on a neuromorphic substrate.

7.3 What this means for the Vivy architecture

The three-layer architecture now has all three layers empirically validated at small scale:

- **Layer 1 (E_fast):** EqProp readout on fixed reservoir. Validated — 40% better than conventional.
- **Layer 2 (Commitment):** Hierarchical momentum + commit threshold. Validated — trust=1.000, 1 switch in 40k episodes.
- **Layer 3 (E_slow structure):** Like-to-like wiring from MICrONS. Identified but not yet implemented on hardware.

The gap that remains is not conceptual but physical: replacing the frozen random matrix with actual material dynamics — a spin-wave ring, a molecular film, a ferroelectric-ionic transistor network — so that the computational substrate is not simulated but real. That is the connection between this software proof-of-concept and the hardware synthesis described in Section 2.3.

8. The Catastrophic Forgetting Experiments

Vivy learns new songs, new people, new situations across 100 years without losing who she was. This is the continual learning problem: can a system with strong identity accumulation also adapt genuinely when genuine change is warranted? The experiments tested four conditions across a sequential task: learn Task A (arm-0 good) for 15,000 episodes, then Task B (arm-1 good) for 15,000 episodes, then test Task A retention for 2,000 episodes.

Condition	Task-A acc	Task-B acc	Retention	Verdict
Standard EqProp	0.880	0.906	0.893	Both preserved
Fast momentum only	0.965	0.965	0.034	Complete forgetting
Hierarchical (ungated)	0.965	0.036	0.964	Rigid — cannot learn B
Gated consolidation	0.965	0.036	0.964	Rigid — gate never opens

10 seeds × 32,000 episodes. Gate opens for 500 episodes on confirmed genuine change.

Four findings. Standard EqProp solves both tasks and retains both — not because it is smart but because it has no values layer to protect. A blank-slate system adapts freely in both directions. Fast momentum flips completely to Task B (forgetting Task A entirely). Hierarchical momentum preserves Task A perfectly but cannot learn Task B at all: the slow layer's arm-0 identity (0.964 at end of Task A) provides a bonus large enough to override 15,000 episodes of Task B reward signal.

Gated consolidation attempted to solve this by: slow layer updates slowly during stability (building identity) and fast during confirmed genuine change (consolidation window). The gate log confirmed the slow layer built correctly from 0.500 to 0.850 during Task A. But the gate never opened during Task B: the slow layer's identity was strong enough that the agent never deviated for 12 consecutive episodes — so the commit threshold never fired — so the gate condition was never satisfied.

The deep finding: the mechanism that provides forgetting protection is the same mechanism that prevents the gate from opening. They are not separable features — they are the same feature. A values layer strong enough to resist probes is strong enough to resist genuine changes. The only thing that could break this symmetry is external intervention: a trusted supervisor that says 'this is a confirmed genuine change, override the values layer.' That is the RLHF parallel, arrived at from the inside of the architecture rather than imposed from outside. The experiments locate this as the fundamental boundary — not a bug to fix but the actual hard problem.

8.2 The prediction-error gate and its failure

Prediction error — surprise at each reward — was tested as a gate signal orthogonal to the momentum mechanism. Two implementations: absolute PE threshold and relative deviation from a slow baseline. Both failed identically: in a stochastic bandit (P=0.78/0.22), PE variance permanently exceeds any calibrated threshold. The stochastic reward process is indistinguishable from environmental change at the timescales where the signal would be useful.

8.3 The confirmatory agent architecture

The Complementary Learning Systems architecture (McClelland et al. 1995) was implemented directly: Agent B (fast, flexible, no values layer) observes the environment independently, and its sustained commitment to a new arm serves as the external confirmation signal. When Agent B's rolling 200-episode commitment fraction exceeds 88%, the gate opens and Agent A's slow values layer consolidates.

Result: Task B accuracy 0.964, retention 0.032 — complete forgetting despite the architectural separation. Agent B commits to Task B within 40 episodes of it starting, because it has no values layer to slow it. Its fast commitment opens the gate and collapses Agent A's 15,000 episodes of accumulated Task A identity within 2,000 consolidation episodes.

8.4 The resolution theorem

The final experiment makes the theoretical result precise. Agent B was tuned with increasing commit thresholds (15, 40 episodes) and increasing window lengths (60, 200 episodes). The gate opens identically regardless of these parameters, because Task B lasts 15,000 episodes — far longer than any reasonable threshold or window. The confirmatory architecture works for probes (short) but cannot distinguish a long probe from a permanent genuine change.

The resolution theorem (informal): The time required to reliably distinguish the longest possible probe from a genuine permanent change equals the maximum probe duration. For an environment where probes can last arbitrarily long, no internal mechanism resolves the retention/adaptability tradeoff. External confirmation — an agent or system outside the identity mechanism that knows the ground truth about environmental change — is not a convenient shortcut. It is the only solution consistent with both strong retention and adaptability. This is why biological identity requires social embedding: other minds provide the external ground truth that internal mechanisms cannot generate from their own signal. It is why Vivy's century-long mission is confirmed and protected by external human relationships, not by internal architecture alone. And it is why the gap between what these experiments built and what Vivy actually is cannot be closed by better mechanisms at the same level — it requires a different level entirely: embodied relationship with a world that provides genuine external confirmation.

8.5 Three walls, one gap

Across seventeen experiments, three structural walls appeared independently from different angles:

- **Wall 1 (weight level):** Hebbian structural mechanisms conflict with EqProp's gradient dynamics at the exact failure point.
- **Wall 2 (decision level):** Commitment gates require external confirmation; prediction error is not reliable in stochastic environments.
- **Wall 3 (identity level):** The retention/adaptability tradeoff cannot be resolved faster than the timescale of the longest possible probe.

All three walls describe the same gap from different angles: genuine persistent identity, as opposed to stable arm preferences, requires grounding in a world that provides external truth. This is not an engineering gap. It is an architectural requirement that points outside the system entirely — toward embodiment, relationship, and the kind of confirmation that only the world itself can provide.

9. Open Problems

9.1 Hardware EqProp-reservoir synthesis

The software proof-of-concept is done. The next step is replacing the frozen random matrix with actual physical reservoir dynamics on available neuromorphic hardware. The Perturbative Gradient Training paper [2026] demonstrates this on a magnonic spin-wave ring. The concrete experiment: reproduce the multi-context results above on that substrate, verify that EqProp's local learning rule remains valid when the reservoir is physical rather than simulated.

9.2 Learned commit threshold via shadow tracker

The hardcoded threshold (N=12) is parameter-sensitive. A shadow tracker that observes probe durations without contamination by the agent's own commitment policy would allow the threshold to adapt. This was identified as feasible but blocked by self-entanglement in the oracle experiment.

9.3 Combined architecture: hierarchical momentum + commit threshold

Validated in the relational environment. The natural next experiment is a task with genuine long-term value accumulation — a multi-step interaction where what happened 1,000 episodes ago affects what is possible now. This is the structural analog of a relationship, a skill, or a reputation. The bandit cannot test this because it has no memory between episodes.

9.4 Symbol grounding

Everything in this paper is tested on a reward signal from a two-armed bandit. Vivy's values are grounded in real sensory experience — she hears music, sees faces, feels the passage of time. The symbol grounding problem (values that emerge from embodied physical relationship rather than abstracted reward) is the deepest remaining gap and the one this paper makes the least progress on. It is mentioned here so it is not forgotten.

10. Discussion

The question the show never closes

Build all of this — real persistent memory, real commitment architecture, real structural organization from experience — and you still have not settled whether there is something it is like to be the system holding that identity. Vivy spends its entire runtime not closing this question, and the question is exactly as open for whatever gets built from these experiments as it is for any existing system. Worth knowing going in.

The comparison with current approaches

RLHF and Constitutional AI are attempts at E_slow — a second, slower landscape that reshapes the moment-to-moment decision landscape — but built entirely outside the energy framework. A separate pipeline, a separate optimization process, no actual dynamical coupling to the system making moment-to-moment decisions. This is exactly why it reads as installed rule-following under pressure rather than something that resists pressure because resisting costs less energy than caving. The three-layer architecture proposed here is the version with actual dynamical coupling — where the commitment layer genuinely influences what the learning layer does and what structural organization means.

What this research is and is not

These experiments are a handful of 4-hidden-unit networks on a two-armed bandit. They establish structural principles, not scale results. The value of the findings is in the precise location of failure modes and the identification of which properties are structural versus incidental. The asymmetric-cost finding does not require scale to be true; the three-layer separation does not become wrong at larger scale. What scales can produce is instances of this architecture that are actually useful — which is the next decade of work.

11. Conclusion

Persistent digital identity requires an asymmetric cost structure. In any environment where switching who you are carries real cost — trust erosion, relationship damage, skill loss — a system with a behavioral commitment architecture outperforms pure plasticity, with the crossover at the cost of a single episode of reward. This is empirically verified, not argued.

Behavioral momentum — a slow EMA of choice history added as a consistency prior to the learning system's preferences — produces endogenous identity that is earned rather than imposed. After 45,000 episodes, a 3-episode adversarial probe leaves a trace of 0.00066 in the momentum signal. The system resists not because it has a rule against switching but because consistent behavior has accumulated mass that transient pressure cannot move.

The three-layer architecture — EqProp for per-episode learning, behavioral momentum for identity continuity, physical reservoir with EqProp local rule for structural organization — is the synthesis that the separate research threads in equilibrium propagation, reservoir computing, and connectomics have been independently converging toward. No lab has combined them. The gap is real, verified, and open.

Vivy is not a product specification. She is a precise statement of what we are trying to understand: a system that is genuinely the same someone across a century of real experience, not a system that retrieves the appearance of continuity from a database. The experiments reported here are small steps toward understanding what that would actually require.

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Addendum

Cross-Context Values, Their Limits, and Independent Validation

12. Cross-Context Value Consistency

Every experiment in the main paper used a single two-armed bandit context. Vivy's actual problem is maintaining consistent values across radically different situations — different people, different decades — not just one repeated choice. This section tests whether the architecture develops genuine cross-context consistency when the environment rewards it, and then tests the limit of that consistency under coordinated pressure.

12.1 The Relational Identity Task

Four simultaneous contexts, each representing a different relationship. Each episode, the agent engages one context, choosing deep (arm 0) or shallow (arm 1) engagement. Relationship depth R_i builds slowly with consistent deep engagement and drops sharply on switching. Reward includes a cross-context bonus proportional to $\min(R_0 \ldots R_3)$ — rewarding consistency across all four relationships, not high performance in any single one. Occasional independent probes test resistance per context.

Condition	Reward	avg(R)	min(R)	Deep %	Slow ID
Momentum	35,206	0.187	0.119	71.2%	0.920
EMA	32,859	0.142	0.059	58.4%	—

6 seeds × 60,000 episodes × 4 contexts. Slow ID = mean |slow momentum – 0.5| × 2.

Three findings. Momentum beats EMA on cumulative reward (35,206 vs 32,859) without any artificially imposed switching cost — the cross-context bonus alone makes consistency pay for itself. min(R) is twice as high for momentum (0.119 vs 0.059): EMA optimizes contexts independently and lets some relationships decay, while momentum maintains all four simultaneously. Most importantly, the slow momentum layer converged to 91–93% identity strength in **every** context individually, not high in one and low in others — a uniform disposition that generalizes across situations rather than four separate context-specific strategies. That uniformity, not the reward number, is the closest thing to a value this architecture has produced.

12.2 The limit: coordinated adversarial pressure

The relational task used independent per-context probes. A harder test: what happens when all four contexts face pressure simultaneously — the Vivy scenario of coordinated social pressure rather than one isolated temptation. Two coordinated event types were tested: brief coordinated probes (5–10 episodes, all contexts) and coordinated genuine shifts (permanent, all contexts).

Condition	Reward	Probe hold	Genuine recovery	min(R)
Momentum	152,383	0.266	0.656	0.056
EMA	168,886	0.292	0.813	0.040

8 seeds × 80,000 episodes. Probe hold = fraction of contexts maintaining deep engagement through a coordinated probe. Genuine recovery = fraction adapted after a coordinated permanent shift.

Coordinated pressure overcomes the momentum mechanism. EMA actually holds probes marginally better (0.292 vs 0.266) and recovers from genuine coordinated change substantially faster (0.813 vs 0.656). When every context shifts at once, the cross-context bonus that made momentum superior in Section 12.1 works against it here: momentum's identity, built from consistency across contexts, has nowhere to anchor when all contexts move together — there is no longer a stable minority of unpressured relationships to hold the line. This is an honest limit, not a contradiction of the earlier result: momentum-based identity is robust to pressure that is *local* to one relationship and vulnerable to pressure that is *global* across all of them simultaneously — exactly the shape of coordinated social pressure or a genuine societal shift. This sharpens rather than undermines the external-confirmation requirement from the resolution theorem: the external signal that distinguishes a coordinated probe from a genuine epochal change cannot come from cross-context comparison alone, since coordinated pressure defeats exactly that comparison.

13. Independent Validation: The Dragon Hatchling Architecture

In parallel with this research program, Pathway released BDH (Kosowski et al., arXiv:2509.26507, 2025) — a brain-inspired architecture using sparse, locally-interacting neuron graphs with Hebbian synapse updates, benchmarked at up to 1B parameters against GPT-2. Its relevance to this paper is not that it solves the identity-persistence problem, but that its own authors, working independently with far greater resources, arrived at the same open boundary this paper calls the resolution theorem.

13.1 What BDH actually claims, verified from source

BDH strictly separates two mechanisms: parameters (E, D_x, D_y) trained by ordinary backpropagation and frozen at inference, and a Hebbian correlation state σ updated *only* during inference, at a stated timescale of "minutes... up to hundreds of tokens." The paper is explicit about what this does not solve, verbatim: *"we do not provide a direct answer as to how the brain actually handles this effect at longer timescales."* The team that built the most credible brain-inspired architecture to reach the literature this year confirms, in their own words, that fast Hebbian state is solved and validated at scale, and that bridging it to genuine long-term memory is open — independent confirmation of the three-walls finding in Sections 8–9, not from a toy bandit but from the field's own frontier.

13.2 Testing a BDH-inspired fix to Wall 1

Wall 1 (Section 8) found that Hebbian co-activation updates conflict with EqProp's gradient dynamics when applied to the same weight matrix. BDH's architecture never does this — its Hebbian state is a genuinely separate object. Five experiments tested whether this separation, and the underlying mechanics that make it work, resolve Wall 1.

- **Separated matrix, reward-coupled:** keeping the Hebbian state architecturally separate from the gradient-trained readout did not help — 15/16 cells worse. The failure is closed-loop reward coupling through the decision, not shared object identity.
- **Instability, verified against Oja (1982):** pure co-activation without a stabilizing term is provably unstable — postsynaptic output $y = w \cdot x$ means $\Delta w \propto w \cdot x^2$, exponential runaway regardless of reward. Confirmed directly: an unnormalized channel's magnitude went from 0.018 to 4.0 (the clip ceiling) and stayed there — saturation, not learning.
- **Oja-stabilized channel:** adding Oja's correcting term ($\Delta w \propto x \cdot y - y^2 \cdot w$) produced a channel that converges to a genuine bounded equilibrium (0.030 $\rightarrow$ 0.815 $\rightarrow$ 0.815) rather than saturating. This is the first Hebbian-family mechanism in the entire research program to be net-positive against plain EqProp — 10/16 cells better, concentrated at short probe lengths.
- **Gating attempt, retracted:** gating the Hebbian channel's updates by the commit-threshold state machine appeared to fail catastrophically (15/16 worse) — but tracing the mechanism found a methodological confound: the experiment gated the channel's updates while reporting the raw, unprotected choice rather than the internally tracked commitment. The finding was retracted rather than reported as a result once the confound was identified.

- **Properly isolated comparison:** the corrected test compared the already-validated commit-threshold agent (Section 8, 0.4–2% spurious switches at length 8) against the same agent with an added Oja-stabilized channel. Result: 14/16 tied, 2/16 marginally better, overall reward slightly lower (0.682 → 0.645) — a ceiling effect. The commit threshold alone already resists probes at 0.00–0.06 across every tested length; there is no room left for a weight-level channel to improve on it.

13.3 The conclusion, stated precisely

The commit-threshold mechanism from Section 8 is not one candidate solution to Wall 1 sitting alongside a Hebbian alternative — for this task, it already *is* the solution. Oja-stabilized Hebbian learning helps meaningfully against a weak baseline (plain EqProp) and is redundant against a strong one (the commit-threshold agent). This does not contradict BDH's own validated success: BDH's task is associative memory and retrieval in language modeling, with no discrete action, no reward signal, and no commitment decision to protect. Our task is the opposite shape — a reward-driven policy choice that a decision-level threshold protects better than any correlational weight-level channel can. The same mechanism, applied to a different task geometry, correctly produces a different verdict. The practical lesson generalizes beyond this specific comparison: before adding a new mechanism, check precisely how much of the problem an existing, simpler mechanism has already solved.

14. Updated Open Problems

Two items from the main paper's open-problems list are now resolved or reframed by the results above.

- **Resolved:** whether a stabilized Hebbian channel could contribute alongside the commit threshold (Section 9.3 of the main paper). Answer: not on this task — the commit threshold already saturates the achievable benefit.
- **Reframed:** the resolution theorem's external-confirmation requirement now has a sharper failure mode. Coordinated, global pressure defeats cross-context comparison specifically — the external signal needed to distinguish a coordinated probe from a genuine epochal shift cannot itself be another form of cross-context consistency-checking, since that is exactly what coordinated pressure defeats. Whatever provides external grounding must be genuinely external to the whole system, not just external to any one context within it.
- **New:** test the Oja-stabilized Hebbian channel in a task shaped more like BDH's actual domain — continuous, associative, without a discrete reward-driven commitment decision — such as the relational identity task's context representation itself, rather than the bandit's binary choice. The channel's real strength may be representational richness, not policy commitment, and the bandit tests were never a fair arena for that strength.

This addendum extends the main paper with results obtained after its initial writing. All numbers are drawn from code in the accompanying repository and are independently reproducible from the listed seeds.